ECON4200 Final Report

The Introduction of Hong Kong's Statutory Minimum Wage: Descriptive Evidence on Changes in Low-Wage Employment by Education and Age Group

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Abstract

This paper documents changes in low wage employment in Hong Kong before and after the 2011 minimum wage policy introduction, focusing on subgroups by education and age. With quarterly microdata from the General Household Survey from 1985 to 2012, we find that there are declines in the proportion earning below HK$28 in both No Schooling/Primary workers and workers aged 50-64. However, robustness tests showed that the decline for No Schooling/Primary workers stands out within the post-1990 baseline, while the decline for older workers is less historically unique.

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1 Introduction

1.1 Policy Background

Hong Kong introduced its first Statutory Minimum Wage (SMW) on 1 May 2011, starting at HK$28 per hour (Labour Department, 2011). This marked a clear institutional turning point, but it is challenging to interpret the before-and-after change of the policy given that the SMW was implemented along with economic recovery, wage increases, and rising rate of inflation. Hence, a simple before-and-after comparison is inadequate for isolating the effects of the policy (HKSAR Government, 2012).

Therefore, this paper takes a more conservative approach. It does not ask whether the SMW caused broad labour market shifts. Instead, the focus is on what happened to the lower end of the wage distribution after the implementation. Previous Hong Kong studies suggest that the outcomes associated with the SMW varied across workers; higher statutory wages did not automatically lead to improved income distribution or employment changes (Lau & Wong, 2018; Ngan, 2022; Song et al., 2026). Thus, this paper asks whether the share of workers earning below some key real hourly wage thresholds decreased after implementation, and whether these changes were larger among low-education workers and older workers.

1.2 Motivation and Research Questions

The motivation is clear and straightforward; to understand whether the low wage shares changed visibly around the introduction of the SMW, especially among workers who are more exposed to lower pay. If a statutory wage floor makes a difference, the most immediate descriptive changes should be found near the lowest end of the distribution. The shares of low wages are more direct; they indicate how many workers remain under that given real-hourly threshold.

This paper will ask three descriptive questions. First, how the share of workers earning below real hourly thresholds of HK$28, HK$35 and HK$40 changed at the initial implementation of the SMW. Second, whether the changes were larger among less-educated workers and older workers. Third, whether the changes around 2011 were historically unusual or whether there are similar decreases in the past, especially in the pre-policy years.

This paper does not take the changes as causal effects as they are patterns in the time span. This paper aims to describe the changes precisely and then compare the changes to historical benchmarks.

1.3 Empirical Approach and Preview of Descriptive Findings

For the initial aggregated analysis, General Household Survey microdata from 1985 to 2012 is used, showing the distribution shifts over time. The analysis gradually directs its focus to the immediate changes before and after the minimum wage policy. Motivated by the discovery of the decreasing proportion of low earners during the policy implementation period, this study conducts a closer analysis of the groups who are more exposed to the minimum wage policy due to a high proportion of individuals with earnings below the minimum wage threshold. After analyzing the trends, older workers aged between 50 and 64, and low-education workers with the classification “No Schooling/Primary” were determined to be the heavily exposed groups under the policy.

This paper then compared the changes on these two groups in 2011 to historical fluctuations (two decades before 2011) to determine the uniqueness of the change in 2011. Robustness tests were also conducted to evaluate whether the changes in these two sub-groups are concentrated around the minimum wage threshold. Results suggested that the shifts in the education subgroup are more unique than those in the age sub-groups, due to the more observable magnitude of change among past fluctuations and the high concentration of change around the minimum wage threshold.

1.4 Value of This Study

There are four ways in which this study contributes. First, it directs the focus towards low-wage shares, as opposed to aggregate wages or employment counts, where the SMW is expected to be most influential. Secondly, it categorizes workers by education and age rather than treating low-wage workers as a single homogenous group. Thirdly, it focuses on quarterly information to concentrate on the immediate window before and after policy implementation. Fourthly, and most importantly, it compares the changes in 2011 to historical fluctuations to critically assess the uniqueness of the shift.

1.5 Report outline

The remainder of the report proceeds as follows. Section 2 covers institutional background on Hong Kong's low-wage labour market, income inequality issues, SMW implementation and the macroeconomic context of periods around 2011. Section 3 offers a literature review. Section 4 and 5 describe the data and methodology. Section 6 discusses the descriptive results and robustness checks. Section 7 outlines limitations, and Section 8 concludes.

2 Institutional Background

2.1 Hong Kong's Low-Wage Labour Market Before 2011

Before 2011, Hong Kong had no statutory wage floor. Wages were determined by the private arrangements and market conditions. The Provisional Minimum Wage Commission (PMWC) described the regime before the policy as "free wage and labour economy" (PMWC, 2010, Ch. 1, para. 1.1).

Low wages were not evenly distributed. They clustered in several sectors: retail, restaurants, and estate management, security and cleaning services, which the PMWC labelled as "Low Paying Sectors" (PMWC, 2010, Ch. 3, para. 3.5). The Commission also identified "older workers" and "workers with low educational attainment" as vulnerable groups (PMWC, 2010, Executive Summary, para. V). This justifies our concentration on subgroups based on education and age, particularly those with no schooling or only primary education, and older workers.

2.2 Income Inequality and the Policy Motivation for SMW

The implementation of the wage floor also happened in the context of increasing inequality. The Census and Statistics Department (C&SD) reported that the Gini coefficient increased, from 2006 to 2011 from 0.533 to 0.537 (C&SD, 2017, Ch. 5, para. 5.18). The problem of inequality of household income was a well-established concern before the implementation of the SMW.

Demographic change added to the complexity. The C&SD reported that the "acceleration of population ageing" and the rise of one-person and two-person households made income differences more pronounced (C&SD, 2017, Executive Summary). In this light, the mentioned official goal of the SMW is carefully articulated to tackle both "forestalling excessively low wages" and "minimizing the loss of low-paid jobs" at the same time (Labour Department, 2011, p. 3).

2.3 Implementation of Statutory Minimum Wage

The SMW came into force on 1 May 2011 at an initial rate of HK$28 per hour (Labour Department, 2011, p. 2). This predetermined rate is central to our empirical analysis as one of our descriptive thresholds is the real hourly wage value of HK$28.

The coverage was extensive as the legislation applied to “all employees, whether they are full-time, part-time or casual employees” (Labour Department, 2011, p. 5). Exemptions applied to live-in domestic workers, student interns, and certain work experience students (Labour Department, 2011, p. 5). At the HK$28 level, the PMWC projected that about 314,600 employees, or 11.3% of Hong Kong employees, would be covered (PMWC, 2010, Ch. 7, para. 7.7).

The timing also shapes our research design. Since the SMW took effect on 1 May 2011, 2011Q2 was a partial implementation quarter. So, this study compares 2011Q1, the last pre-policy quarter, with 2011Q3, the first quarter after implementation.

2.4 Macroeconomic Context Around Implementation

The SMW did not land in a stable economy. Hong Kong was recovering from the global financial crisis when the policy was introduced. The 2011 Economic Background Report notes that the economy expanded at a 7.0% growth rate in 2010 and continued its growth with a "notable 5.0% expansion" in 2011 (HKSAR Government, 2012, p. 1).

Labour market conditions were also tight with the seasonally adjusted unemployment rate dropped to its 13-year low levels of 3.2% in the third quarter of 2011 (HKSAR Government, 2012, p. 1). Prices were also going up as underlying inflation of 2011 was 5.3%, up from 1.7% in 2010 (HKSAR Government, 2012, p. 2).

This macroeconomic context is important for the analysis, as this study used real hourly wages to see if workers earned above key low-wage thresholds, adjusting for inflation. Thus, the descriptive changes discussed later in this paper should be interpreted within the context of economic recovery, tight labour conditions, and escalating prices.

3 Literature review

3.1 International Evidence and the Distributional Lens

Minimum wage debates often begin with the question: what happens to employment after the wage floor is imposed? Card and Krueger (1994) took this question to New Jersey and Pennsylvania. After New Jersey raised its minimum wage in 1992, they examined the fast-food industry and found that employment did not decrease; rather, it increased slightly compared to Pennsylvania. This result was controversial. Neumark and Wascher (2000) analyzed the same policy change using payroll records, but their findings indicate a modest employment decline once measurement problems were addressed.

What this paper can apply from their findings is that minimum wage results are sensitive to data sources and methodologies; the same case could obtain different findings. That sensitivity should be noted for Hong Kong studies, as our data do not support a clean causal analysis. So, this paper chose a conservative approach: a descriptive focus on wage distribution, rather than a strong causal argument about employment (Card & Krueger, 1994; Neumark & Wascher, 2000).

Related studies shift focus from employment to the lower tail of the wage distribution. Autor et al. (2016) show that the minimum wage led to changes in lower-tail inequality in the US, particularly among women and low-paid workers. This paper shares that focus on distribution. But Autor et al. (2016) also warn that apparent spillovers above the minimum might reflect only measurement errors; not necessarily real pay increases. That warning applied to our study because hourly wages in this analysis are constructed from banded monthly earnings and weekly hours. Thus, careful descriptive measurement at the lower tail is a more suitable approach.

3.2 Hong Kong Evidence and Vulnerable Workers

Hong Kong's statutory minimum wage provides a different institutional context. Song et al. (2026) use the General Household Survey from 2011 to 2019 to investigate five cycles of SMW changes. They found no systematic rise in underemployment or reduction in working hours, but the wage gains among low-income workers were unevenly redistributed by sectors and cycles. This paper does not replicate their causal design but adopt an important idea: Wage outcomes in Hong Kong are not homogenous and aggregate results may hide important heterogeneity among different groups (Song et al., 2026).

Other studies of Hong Kong all point in a similar direction. Ngan (2022) argues that a higher statutory wage does not necessarily translate into real poverty reduction, when living costs increase faster. Lau and Wong (2018) similarly show higher pay may be absorbed by household costs, such as food, housing and other household expenses. Wong and Ye (2015) found positive wage and income effects on vulnerable groups, but no clear evidence on substantial employment losses.

Hon (2015) looked at security companies and found that although wage floor reduced pay gaps, some firms responded by moving some employees toward part-time work. That complicates the picture. The SMW may be associated with low-end pay, but what actually happens to workers depends on hours, living costs, how firms adjust, and which sector they work in (Hon, 2015; Lau & Wong, 2018; Ngan, 2022; Wong & Ye, 2015).

Tan and Ko (2010) noted that the exposure to the wage floor would be uneven across workers; low-wage sectors, such as retail, restaurants, security and cleaning services would be affected most directly by the SMW. This suggests that it is reasonable for this paper to examine low-wage shares separately by education and age, rather than putting all workers into an aggregate average.

3.3 Research Gap and Contribution

Taken together, the international and Hong Kong literature points to a common caution: minimum wage outcomes are sensitive to data, method, and context. International studies show that different methodologies yield different findings. Hong Kong studies suggest that low-end pay may rise around the SMW, but unevenly across time, sectors, and worker groups. The 2011 implementation took place amidst the recovery from the global financial crisis, so it is hard to separate policy effects from broader economic changes. For this reason, this paper does not treat the SMW as a clean causal shock. Instead, it documents what changed implementation and asks whether those changes were unusual relative to historical fluctuations. Following Autor et al. (2016), who emphasize the lower tail of the wage distribution, and Song et al. (2026), who find that Hong Kong's SMW may differ in strength across time and groups, the results section focuses on the share of workers earning below real hourly thresholds: HK$28, HK$35, HK$40, not just average wages. That gives a clearer descriptive picture of the lower tail, but does not prove causation.

Some research gaps remain in the Hong Kong literature. Most studies do not put the 2011 implementation within a long historical baseline. Some studies investigate only the period after the implementation of the policy, while others include the period just before the implementation. They leave a question to examine; whether the improvements to some groups of workers in 2011 are historically unusual or just a feature of normal fluctuations.

This gap is where this paper contributes; it compares the 2011 statutory minimum wage period against historical fluctuations. This enables the interpretation of the observed changes within a longer period of context. This paper is not trying to prove causality as it would require a sophisticated methodology. Instead, the intention is to describe the timing, size, and subgroup pattern of the change and compare them with previous historical changes. The goal is modest but useful: to assess whether the immediate changes around the 2011 implementation were historically distinctive, and to identify which groups that experienced the largest changes (Song et al., 2026; Ngan, 2022; Tan & Ko, 2010).

4 Data and Methodology

4.1 Data source

We will be using the General Household Survey microdata. It is a monthly survey conducted by the Census and Statistics Department (C&SD) with variables including but not limited to gender, industry, and wage. The mappings of key variables are provided in Appendix A.

4.2 Hourly Wage Construction

We use midpoint estimations to derive monthly earnings. As we have noticed, the monthly earnings were given as ranges as shown in Appendix B. For example, 01 stands for an earning range between 0 and 2000 and 02 represents a band between 2000 and 3000. Since they are banded, we will be using midpoint approximation. For instance, an individual earning the range 2000-2999 will be set to 2500.

There exist estimation errors when setting every individual's earnings in a wage band to the midpoint value. Thus, robustness checks will be applied in the following discussions.

There exist challenges in deriving the hourly wage because the data only provides us with weekly working hours and monthly salary. To estimate a person's monthly hour, we decided to multiply this weekly working hour by the number of weeks (365 divided by 7) to obtain the total annual working hours. Then the monthly working hours can be derived by dividing the annual hour by 12. Intuitively, the hourly wage could be obtained by dividing the monthly wage by the monthly working hours.

$$Hourly\,Wage\;=\;\frac{Monthly\,Salary}{Monthly\,Hours}$$

$$=\frac{Monthly\,Salary}{Weekly\,Hours\;\times\;Number\,of\,Weeks\,Per\,Month}$$

$$=\frac{Monthly\,Salary}{Weekly\,Hours\;\times\;(365\div 7)\div 12}$$

To obtain their real wages, we will be using the Consumer Price Index (CPI) obtained from the Census and Statistics Department, which gives us the monthly CPI from 1975 to up to 2026 March. We will only be using the 1985 to 2012 ranges. Since we are looking into the data by quarter, we will obtain the quarterly average CPI by averaging the CPI of each constituent month in that quarter. With the CPIs obtained, the real hourly wage could be obtained through the following. In this case, we take the CPI in 2011 quarter 2 as reference, which is 78.

$$Real\,Hourly\,Wage\;=\;\frac{Hourly\,Wage}{Current\,CPI}\times\;Reference\,CPI$$

4.3 Education Recoding

For educational coding, there was a change in classification method in 2012 Q1, making the terminologies slightly different. As shown below, code 1 to 4 remained unchanged whereas code 5 (Six form), code 6 (Technical/vocational Training) and code 7 (non-degree courses) were shifted into the categories of diploma, certificate and sub-degree in 2012. Code 8 (first-degree courses) and code 9 (post-degree courses) remained with a mild change in the names.

From Q1 1985 to Q4 2011:

Code	Description
1	No schooling / pre-primary
2	Primary
3	Lower secondary
4	Upper secondary
5	Six form
6	Technical / vocational training
7	Tertiary (non-degree courses)
8	Tertiary (First degree courses)
9	Tertiary (Post-degree courses)

From Q1 2012 to current:

Code	Description
1	No schooling / pre-primary
2	Primary
3	Lower secondary
4	Upper secondary
5	Post-secondary – diploma / certificate including craft
6	Post-secondary – sub-degree
7	Post-secondary – first degree
8	Post-secondary – post-degree

To deal with this discrepancy, for the data before the 2012 shift, we keep all categories. For the data after the shift, we retain code 1-4 and 7-8 as those classifications were unchanged. We discard the data categorized as diploma, certificate, or sub-degree after 2012 Q1 due to the lack of consistency with previous data. This is the reason why, in some diagrams in the following discussions, the lines showing the trend of code 4-6 abruptly stopped in 2011.

4.4 Age Grouping

Similarly, different age groups have different age code:

code	description
01	<5
02	5 - 9
03	10 - 14
04	15 – 19
05	20 – 24
06	25 – 29
07	30 – 34
08	35 – 39
09	40 – 44
10	45 – 49
11	50 – 54
12	55 – 59
13	60 – 64
14	Equal to and older than 65

We have classified them into 3 categories: young workers (15-29), prime-age workers (30-49), and older workers (50-64). Young working age groups likely consist of students and citizens who are in their early careers. Old working age groups are likely to have less well-paid positions such as sanitary workers or security guards. Thus, young and older workers are expected to have more exposure to the policy, as shown in the following analysis.

4.5 Grossing up factor

The General Household Survey includes a grossing up factor variable, which inflates sample observations to population totals. We use this as a sampling weight to ensure our estimates are representative of Hong Kong's employed population. All reported proportions and means are weighted using this factor.

4.6 Analytical Approach

This paper employs a purely descriptive empirical strategy. We document changes in low-wage employment that coincided with the implementation of the Statutory Minimum Wage (SMW) in May 2011.

For both the education and age subgroup analyses, we focus on the 2002 to 2012 period. The reason to exclude pre-2002 period is because the proportion of workers earning below HK$28 was high. Including these years could overstate any coincident changes from the policy.

Our analysis proceeds from aggregate level to more specific analysis on different groups. First, we analyze the trend in wage distribution shifts for the entire population, ignoring factors such as ages and education levels. This bird's-eye view enables us to identify the trends before and after the implementation of minimum wage policy. Following this, we would have a deeper exploration into the groups which contributed the most to those shifts. Specific investigations will be into the age groups and education groups who are likely to be heavily affected by the minimum wage policy.

To avoid potential smoothing from multi-quarter averaging, we compare the immediate pre-policy quarter (2011Q1) and the first full post-policy quarter (2011Q3). We exclude 2011Q2 because the policy took effect on May 1, making this quarter a partial treatment period. This comparison isolates the most immediate change around the policy implementation date.

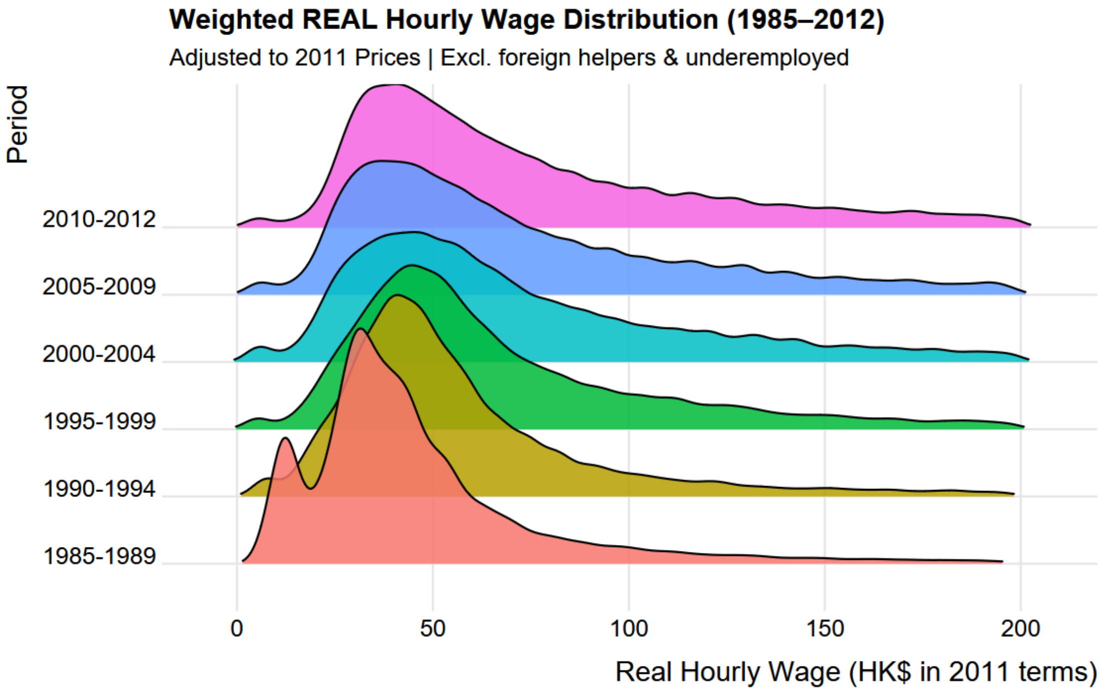
We assess whether the observed 2011 decline was historically unusual by conducting a series of robustness checks. We calculate the year-over-year declines in the proportion of people earning lower than HK$28 per hour from 1986 to 2010, which serves as a benchmark to determine whether the decline in 2011 was unique. We also analyze the short-term fluctuations without the policy impact by calculating the quarterly shifts in income distributions. Following this, another robustness test which changes the threshold of focus will be conducted to analyze whether the decrease in proportion is uniquely concentrated on the minimum wage threshold.

5 Results

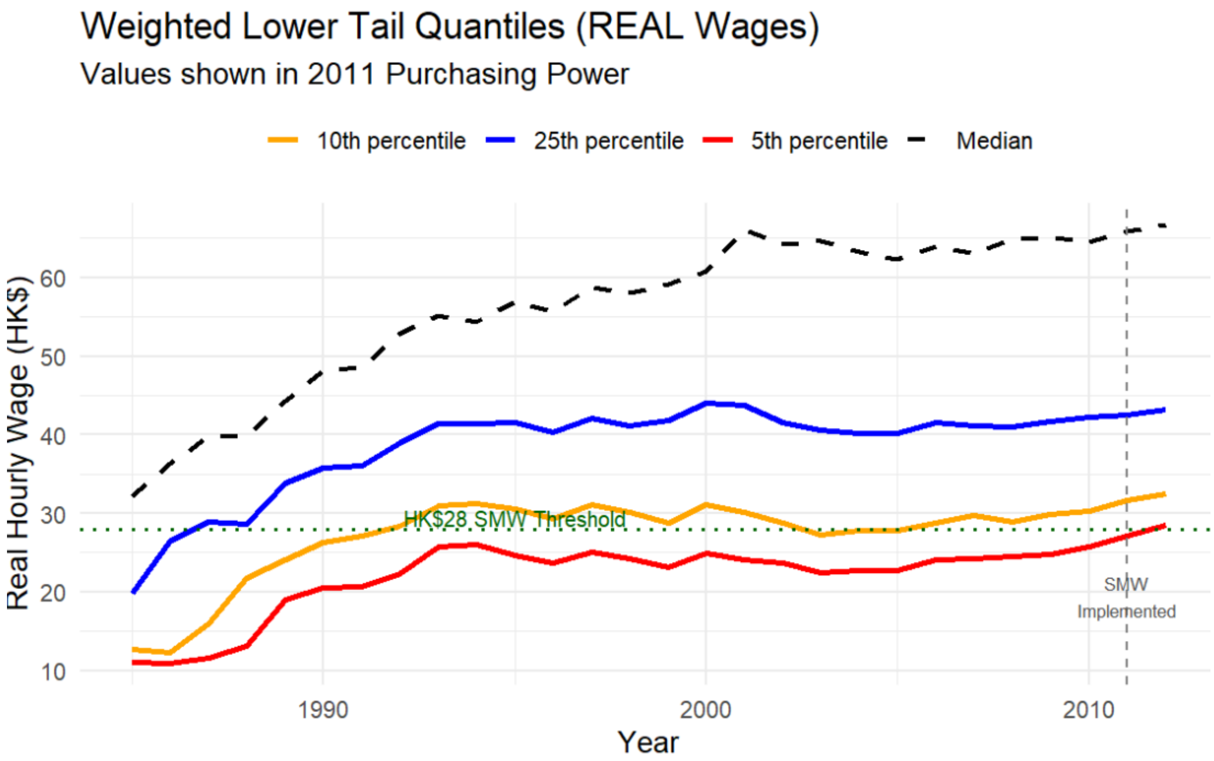
5.1 Aggregate trends

In this section, a more macro-level analysis will be applied to Hong Kong labour force, which considers people of all working ages and education levels and wages. The analysis gradually narrows its focus to the relatively low-income groups and a specific time interval where important policy was implemented.

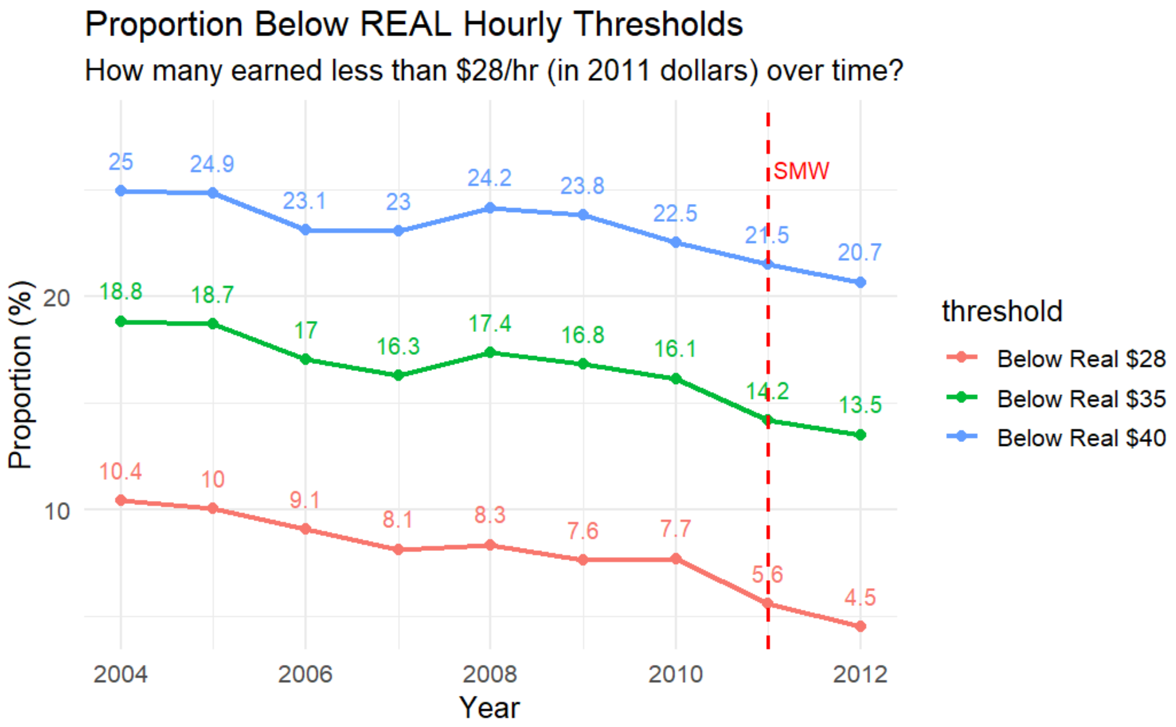
After filtering out the individuals who are out of working age or underemployed and excluding the ones who are foreign helpers or have missing data, a density diagram of real wages was plotted across a time span from 1985 to 2012. As shown below, from 1985 to 1989, the distribution was concentrated at low wages, peaking at HK$20 to 30 per hour. Over time, the distribution shifted moderately to the right. From 2010 to 2012, the peak had moved to approximately HK$30 to 50 per hour. The left tail, which represents low-income groups, became noticeably thinner compared to the 1985-1989 period. This suggests a gradual decline in the proportion of workers who earn low wages.



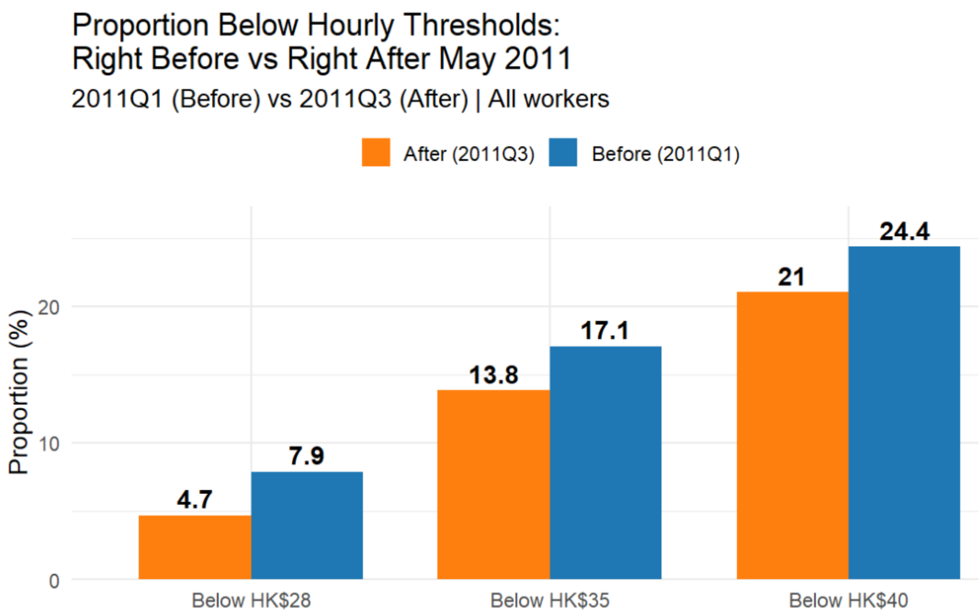
Looking specifically at the low-income groups, as shown in the line diagram below, which illustrates the hourly income at low percentiles, the 5th percentile real hourly wage increased from around HK$24 to HK$28 from 2003 to 2011, and the timing of this increase coincides with the minimum wage implementation in May 2011.



To investigate the changes in proportion in the population of low-income groups, a line diagram showing the proportion of citizens with real hourly income lower than various thresholds is shown below. The graph reveals a downward trend in the proportion of people earning relatively lower real hourly income. More importantly, the proportion of low-income group earnings below the minimum wage threshold (HK$28) decreased by more percentage points relative to other thresholds in 2011. This decline coincides with the implementation of minimum wage policy in May 2011.



To see if there are any noteworthy changes before and after the minimum wage policy, a bar chart below compares the quarter immediately before policy implementation (2011 Q1) and the quarter immediately after (2011 Q3). The proportion of workers earning below HK$28 declined from 7.9% to 4.7% over this period.

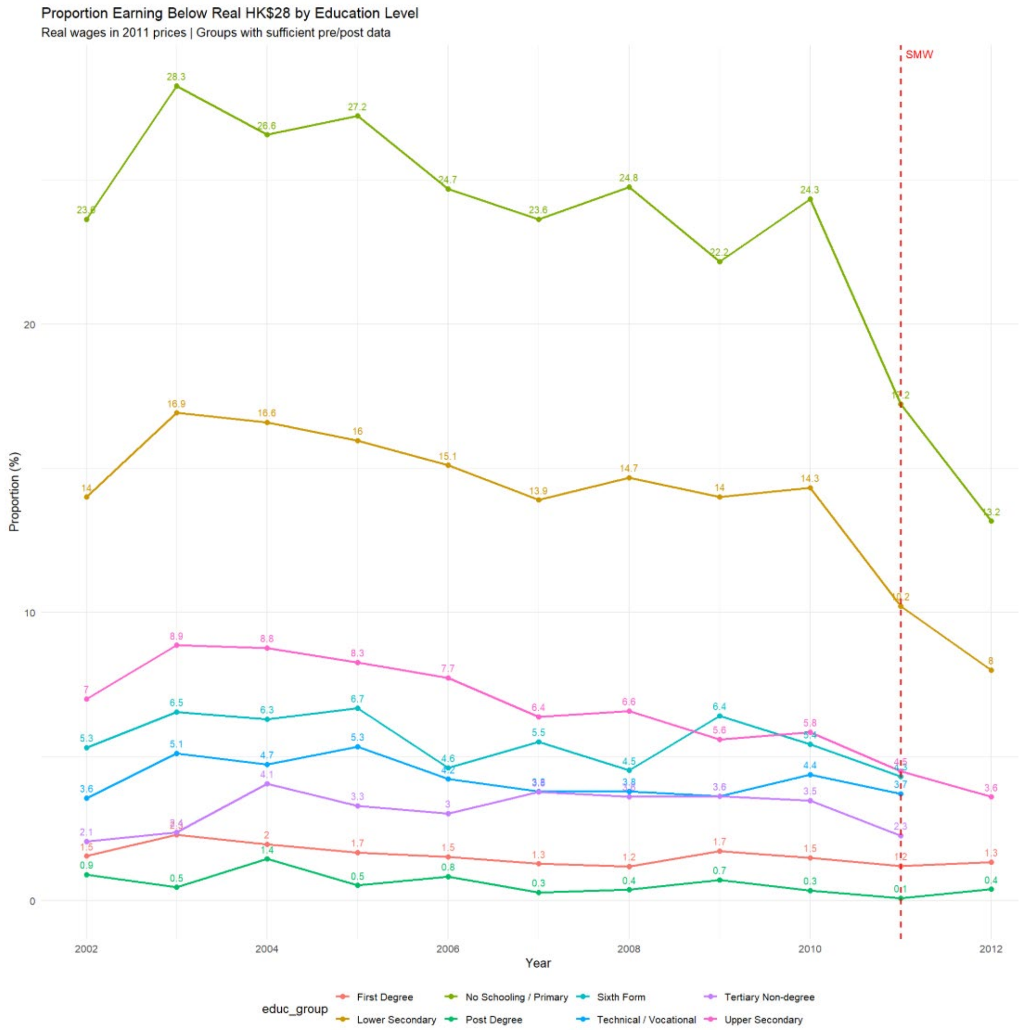


In summary, there was a noticeable improvement in the wage distribution of Hong Kong labour force during the period of minimum wage implementation, with a decline in proportion of workers earning below HK$28 per hour. In the following discussions, sub-groups who have more proportion of workers below the minimum wage threshold will be analyzed to potentially explore the impact of the policy.

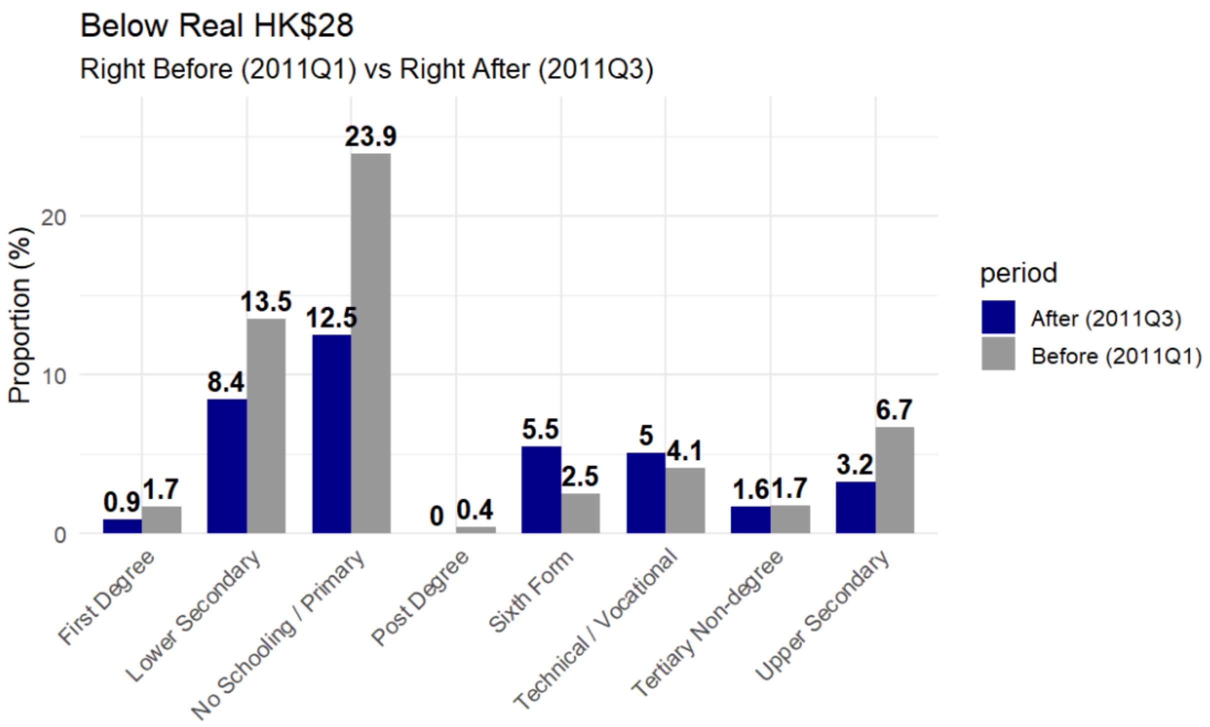
5.2 Education Subgroup Analysis

In this section, changes in proportion of low-income workers with different education backgrounds will be analyzed to identify the most exposed education group. For the group with the largest decline during policy period, its year-on-year changes are compared to historical fluctuations to see whether the change during the minimum wage period was unique.

The line diagram below shows the change in proportion of workers with real hourly earnings below the minimum wage threshold for various education levels. Due to the change in classification methods in 2012Q1, education levels such as “Six Form” are stopped in 2011. From the line diagram, it is apparent that workers with education levels of “No Schooling”, “Primary” and “Lower Secondary” experienced a larger decline in their low-wage proportion compared to other education levels. In addition, these education groups also had the highest pre-policy proportion of people below the minimum wage threshold, making them the most exposed group under the policy.

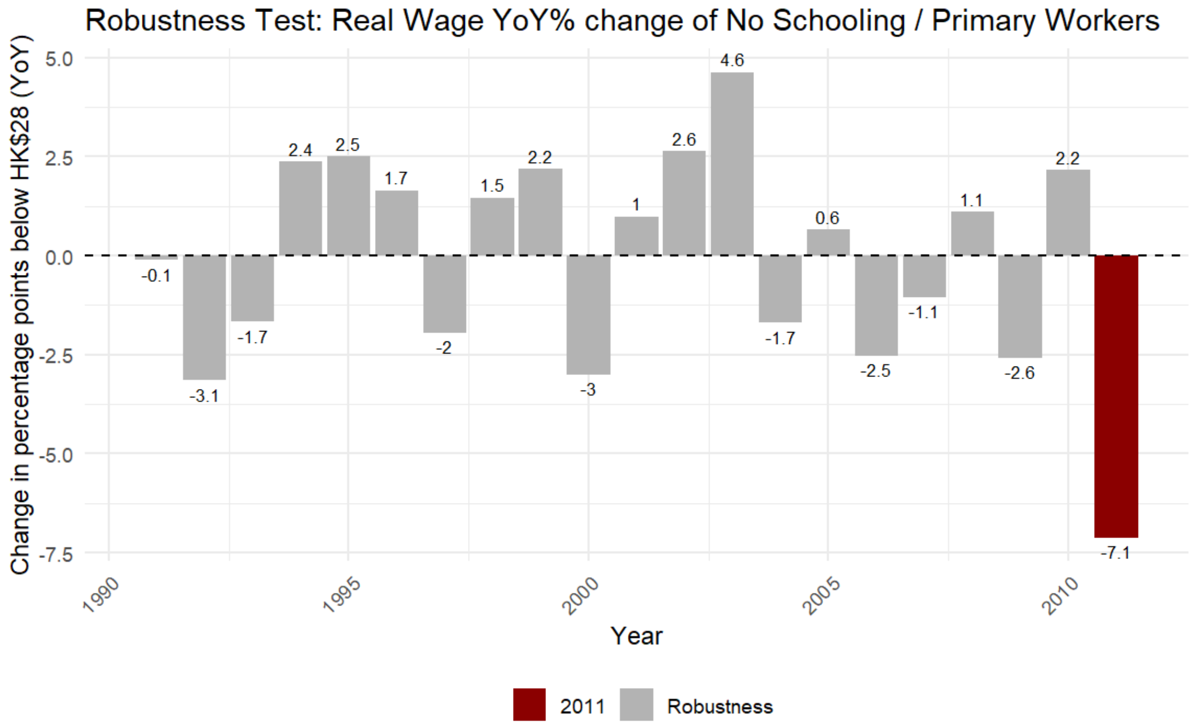


Looking at the quarterly data immediately before and after the policy, consistent findings could be derived as the workers with education level of "No Schooling/Primary" who were earning below HK$28 per hour declined from 23.9% in 2011Q1 to 12.5% in 2011Q3, which is a reduction of 11.4%.



It is difficult to determine the net policy impact on the aforementioned education group as those changes could be driven by various factors such as economic cycles. To assess whether the 2011 decline was historically unusual within the 1990-2010 baseline, robustness checks are applied as follows.

As shown in the bar chart below, which illustrates the year-on-year change in percentage point of people with education level of "No Schooling/Primary" who are earning below the minimum wage threshold, the change in proportion in 2011 was the highest in the two decades before 2011. Most of the change in percentage point was around 2-3 in terms of absolute value, whereas the change in 2011 reached 7.1, standing out from previous fluctuations. Thus, the 2011 decline stands out as larger than any year-over-year change observed for this group in the two decades prior to 2011.

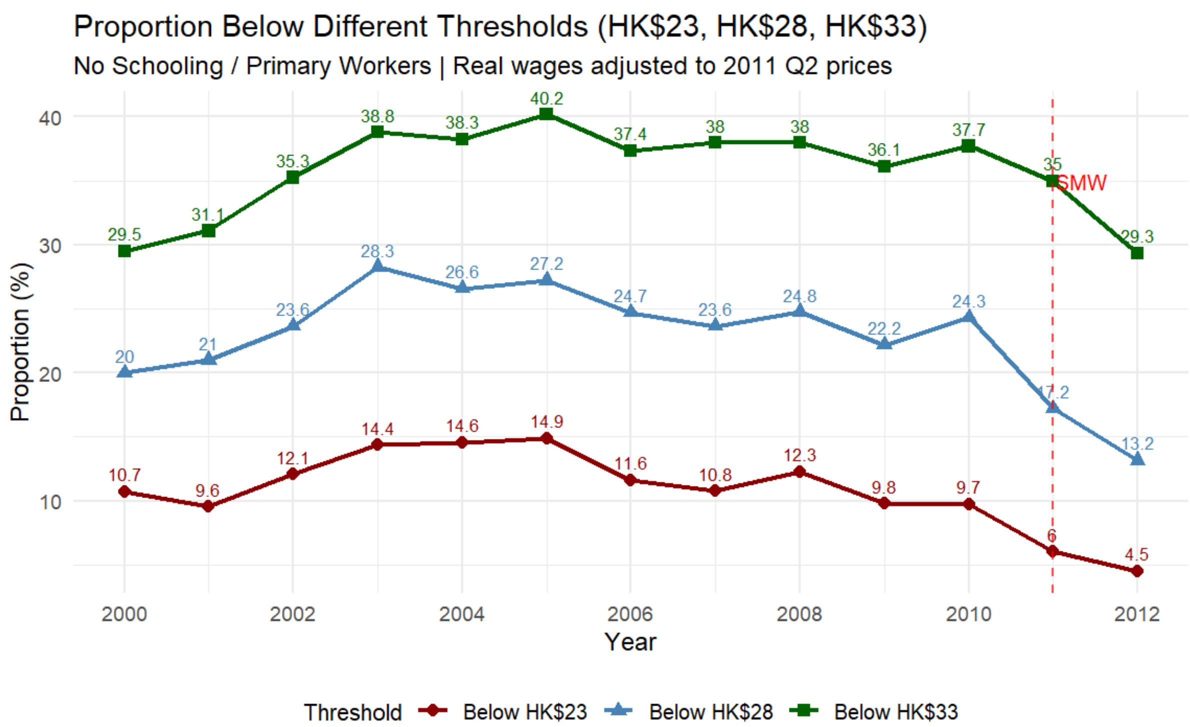


For this analysis, the data before 1990 will be filtered. The main reason is that multiple sources stated that Hong Kong underwent structural economic changes during the 1980s. For example, Chiu, Ho, and Lui (1997) documented that this period is marked as a fundamental industrial restructuring. There can be large changes in the proportion of wage distribution in this period, which is not comparable to stable periods during the minimum wage implementation. Therefore, a more conservative baseline will be only considering 1990 to 2012.

A different but related question is whether the decline was concentrated near the statutory minimum wage threshold. To test this, a few thresholds were applied as shown in the line diagram below, which are HK$23 (HK$5 below SMW), HK$28 (the SMW level), and HK$33 (HK$5 above SMW).

Between 2010 and 2011, the proportion below HK$28 declined by 7.1 percentage points (from

24.3% to 17.2%). The change for the HK$23 threshold was 3.7 percentage points (from 9.7% to 6.0%), and the decline for the HK$33 threshold was 2.7 percentage points (from 37.7% to 35.0%). The largest decline occurs exactly at the statutory minimum wage threshold, which is observably larger than other thresholds.

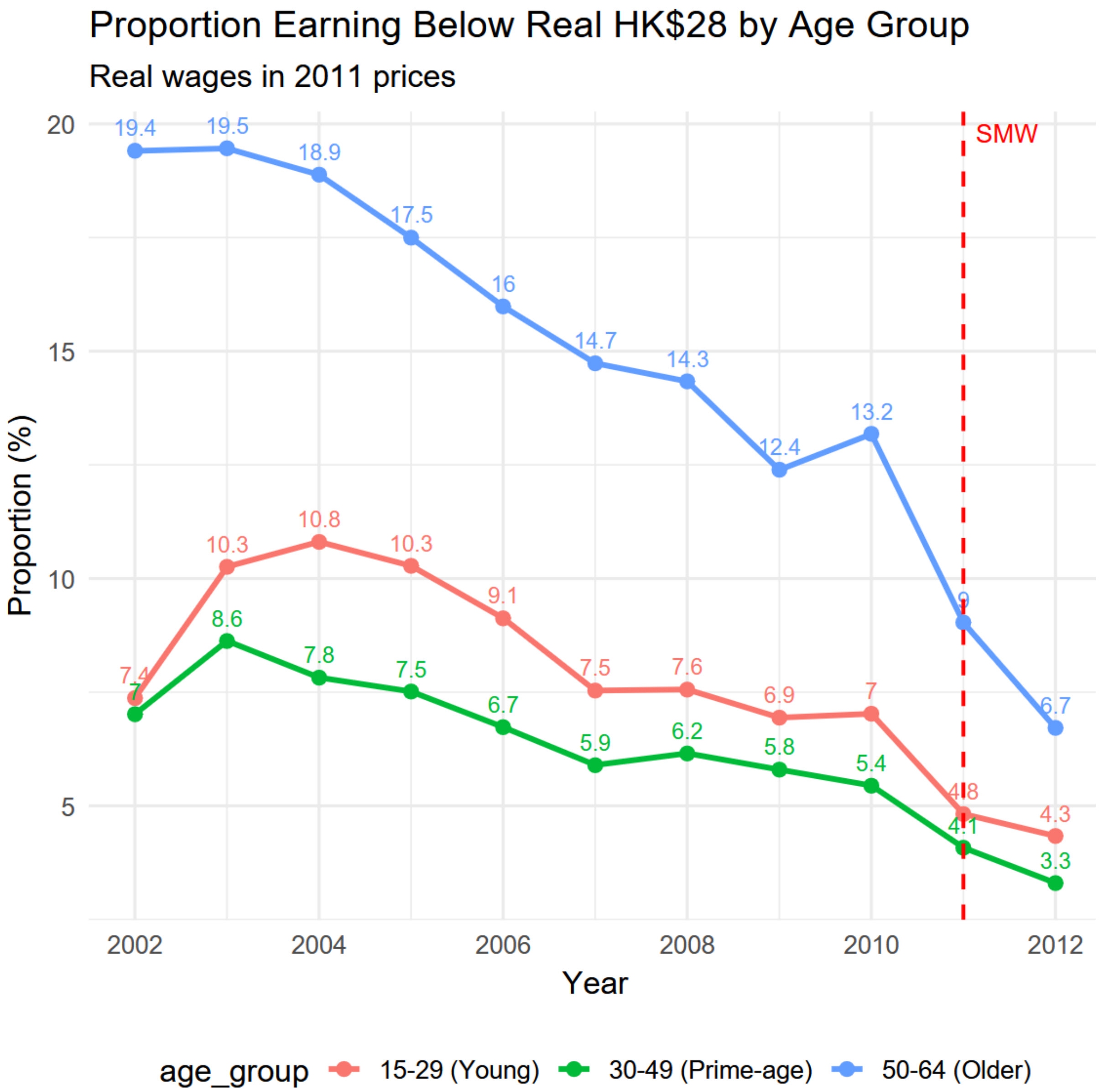


This shows that for No Schooling/Primary workers, the 2011 decline was historically unique in magnitude, and was concentrated at the policy threshold.

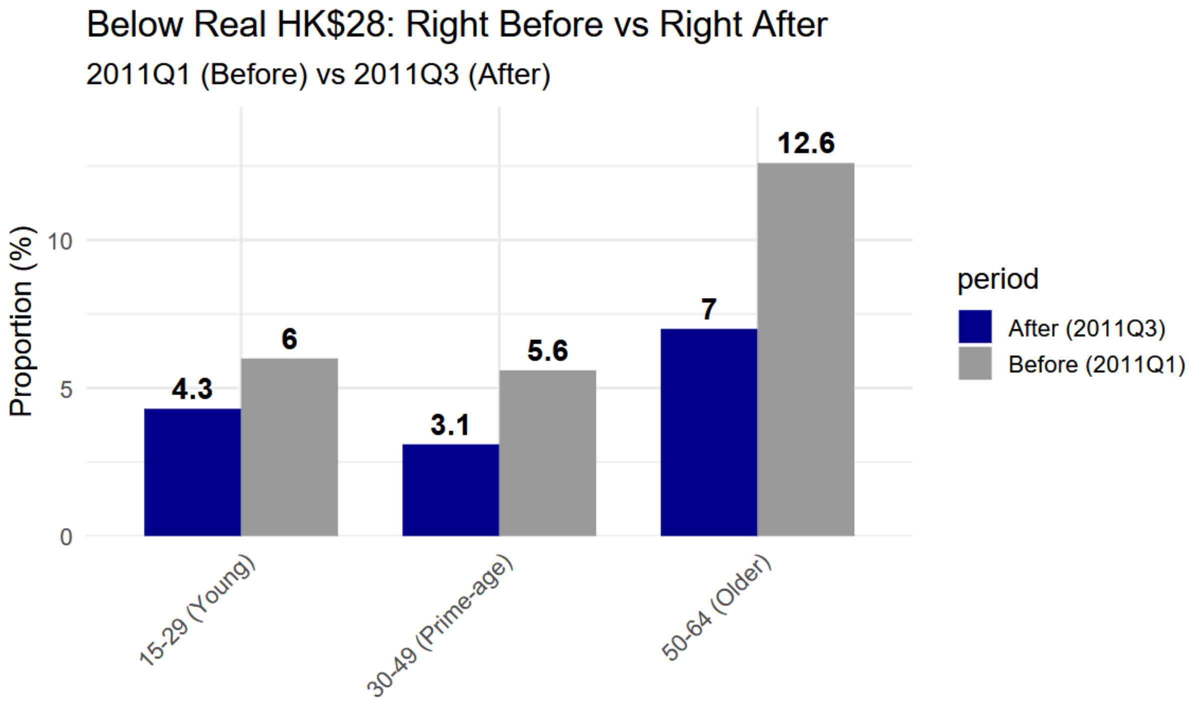
5.3 Age Subgroup Analysis

Similar to previous procedures, for various age groups, the shift in proportions of individuals below minimum wage threshold will be analyzed, followed by a specific analysis to the groups who are likely to be heavily exposed under the policy.

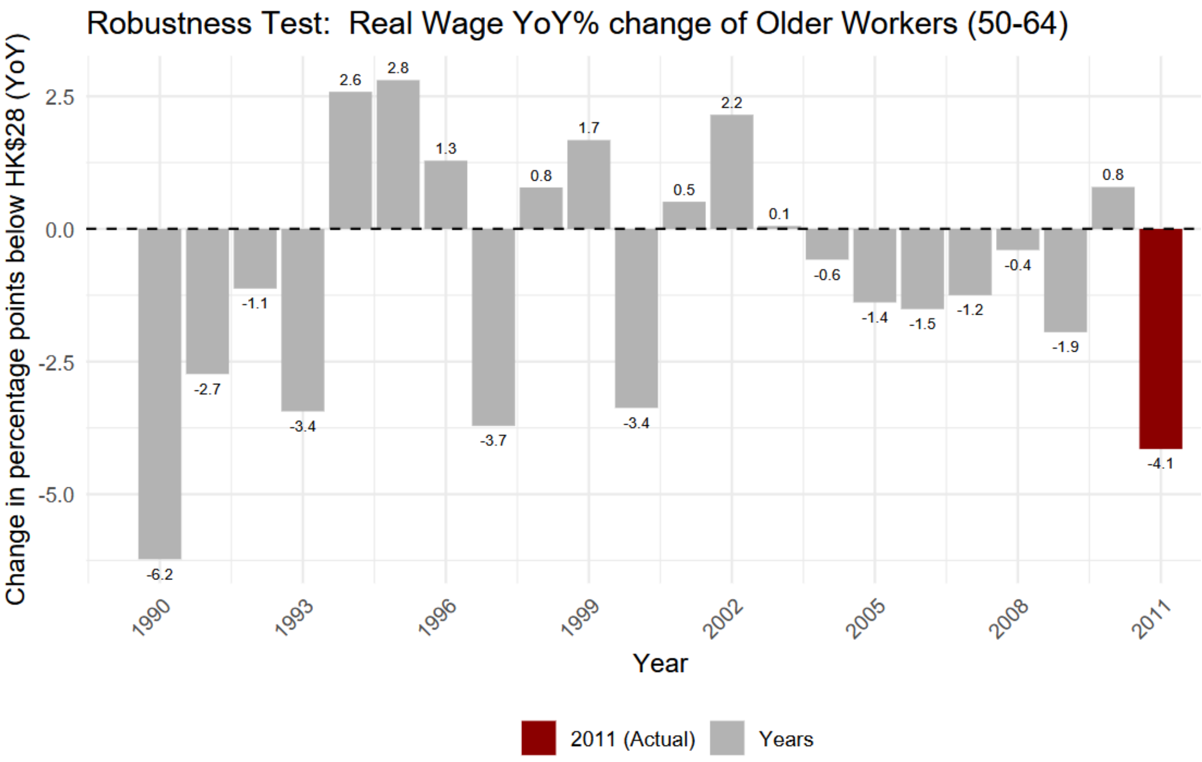
As shown in the line diagram below, there is a declining trend in proportion of people below minimum wage threshold across all age groups. The decline of 50-64-year-old workers earning less than HK$28 was the sharpest, plummeting by 4.2 percentage points from 13.2% to 9%. Meanwhile, this age group has the most proportion of people below the minimum wage threshold, making them the most exposed group to the policy.



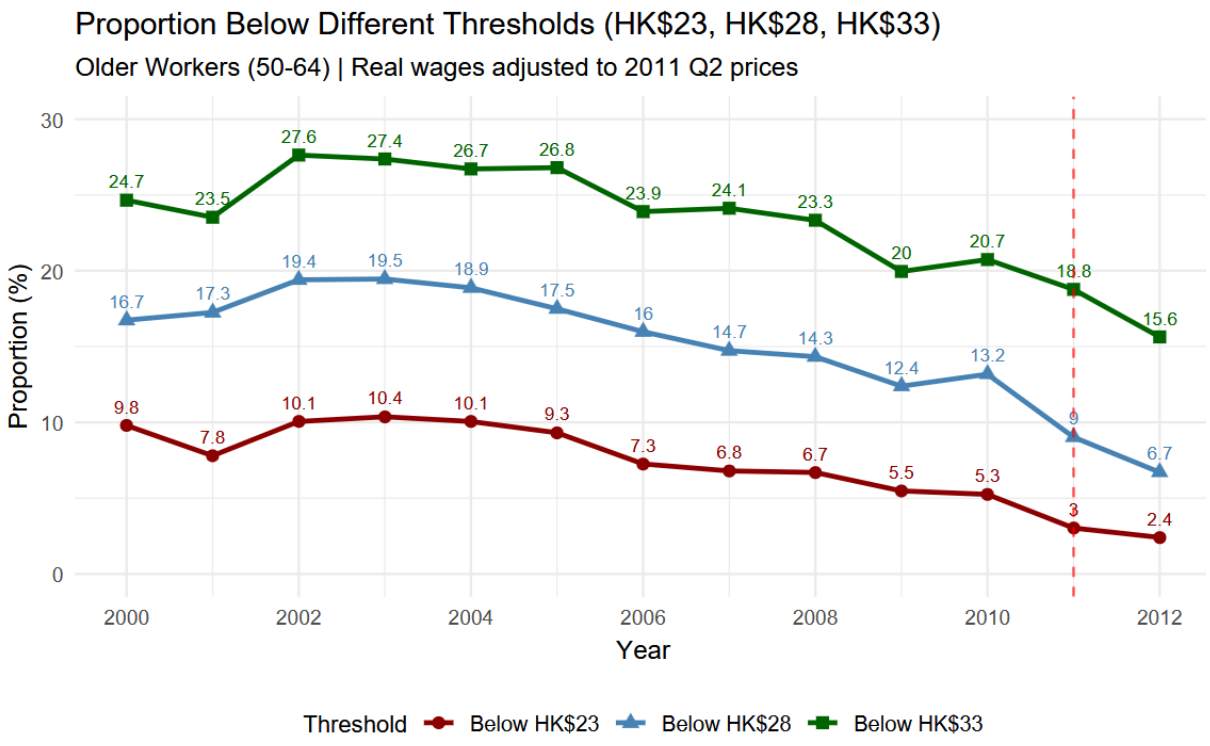
Like previous approaches, a bar chart showing the quarters right before and after the policy is sketched below, suggesting consistent finding with previous guesses. The proportion of individuals earning below the minimum wage threshold who are in the age interval of 50-64 experienced a slump from 12.6% to 7%. This change is remarkably larger than other age groups.



To assess whether the 2011 decline was historically unusual, the following robustness checks are conducted. The bar chart below shows the year-on-year change in percentage point of proportion of workers earning below the minimum wage threshold who are aged 50-64. It could be determined that the change in low-income groups' proportions in 2011 for older working age workers was less unique for two reasons. First, there were drops of similar sizes in the years 1997 and 2000. Second, from 2005 to 2009, there has been a pre-existing downward trend with consistently negative change in percentage points. Although the sizes were smaller than the change in 2011, this suggests that pre-policy trends may have contributed to the decrease in percentage points in 2011.



Historically, the 2011 decline for older workers was less unique in magnitude. Similarly, the 3 thresholds (HK$23, HK$28, and HK$33) will be tested again on the age group 50-64 to see if the decline was concentrated near the minimum wage threshold. As shown in the line diagram below, between 2010 and 2011, the proportion below HK$28 declined by 4.2 percentage points (from 13.2% to 9%). The proportion for the threshold HK$23 declined by 2.3 percentage points (from 5.3% to 3%). The change for the threshold HK$33 is -1.9 percentage points, (from 20.7% to 18.8%).



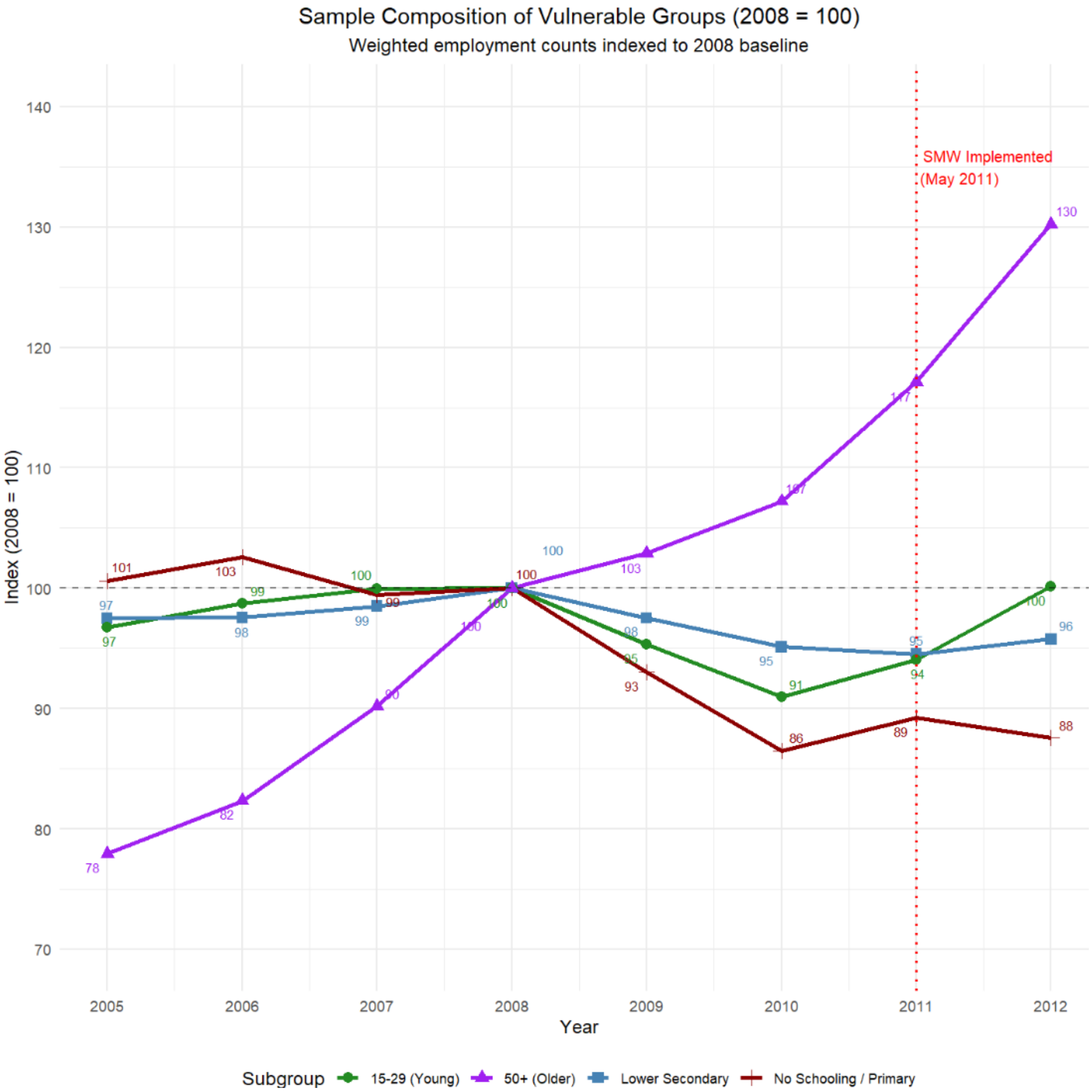
For older workers, the decline was less concentrated at the minimum wage threshold compared to the low-education groups, which is consistent with the pattern found earlier with 2011 decline for older workers being historically less unique.

5.4 Additional Robustness Checks

This section focuses on the concerns of estimation errors and other factors which might explain the proportion decrease shown before.

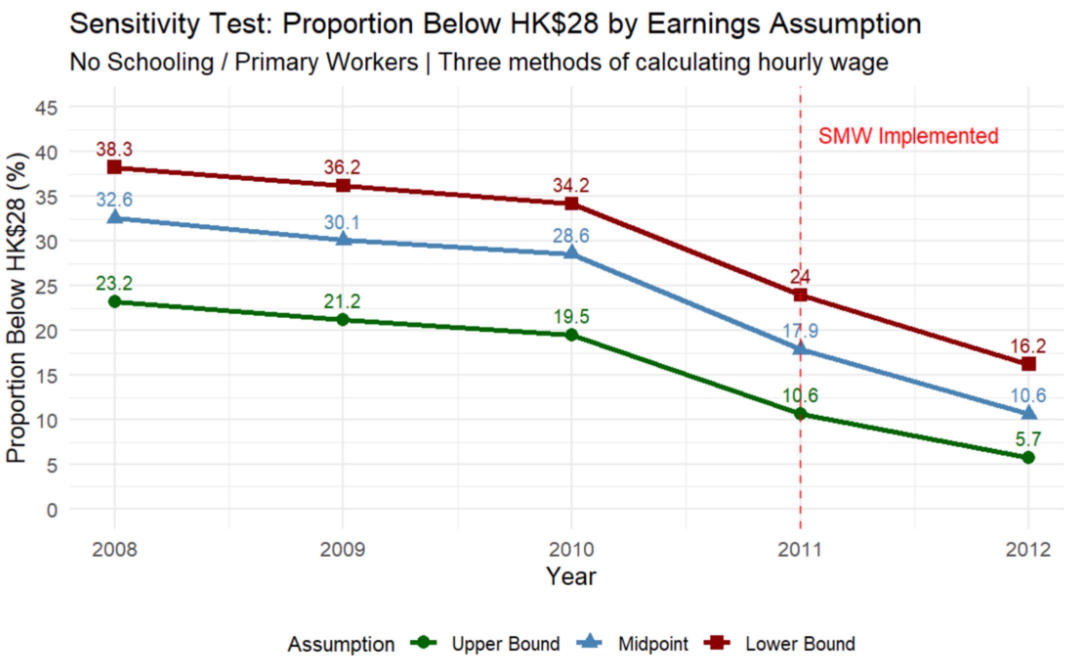
First, as elaborated in the previous discussions, there was a decrease in the proportion of workers earning below the HK$28 threshold, which was attributed to the improvements in income distribution. However, there is a possibility of a decrease in employment of low-income groups, which reduces the proportion of workers earning below HK$28 per hour. To analyze the impact of employment change on the shifts in proportion, the line diagram showing the employment counts for various groups involved in the previous discussion is sketched, with the employment counts in 2008 taken as the base year.

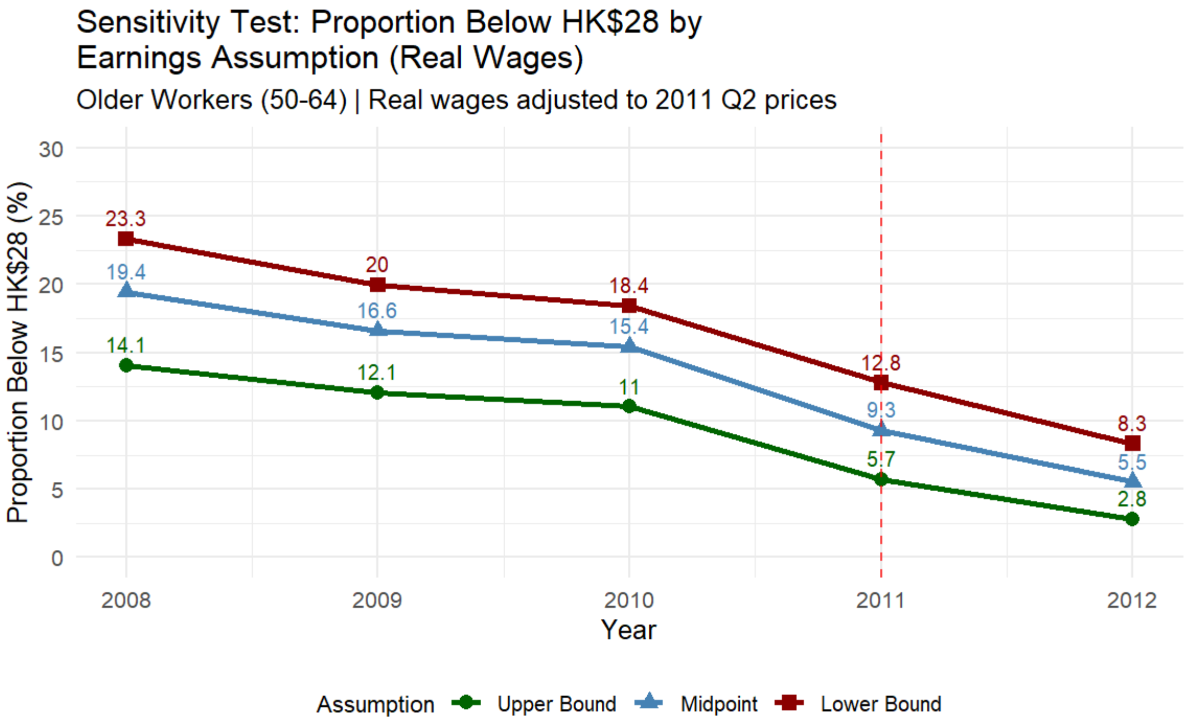
Through the diagram, a consistent rising trend in employment counts could be identified across various heavily exposed groups from 2010 to 2012. This suggests that the proportion shifts are likely not driven by low-income groups being excluded from the sample in 2011. It could be suggested that the proportion change was due to the increasing number of individuals who experienced a rise in earning to higher income intervals.



In addition to the sample composition concern, it is also possible that using midpoint approximation could lead to a large measurement error. As a result, a sensitivity test on proportion of workers earning below HK$28 per hour was drawn using 3 types of approximation: lower bound, upper bound, and midpoint. For example, for an earning band between 2000 and 2999, the upper bound, midpoint and lower bound will be 2999, 2500, and

2000 respectively. As shown on the graph below, the trend using different approximation has been largely parallel, meaning that the finding of the paper is not sensitive to the choice of approximation methods.




Sensitivity Test: Proportion Below HK$28 by
Earnings Assumption (Real Wages)
Older Workers (50-64) | Real wages adjusted to 2011 Q2 prices
Proportion Below HK$28 (%)
30
25
20
15
10
5
0
23.3
20
18.4
12.8
8.3
19.4
16.6
15.4
9.3
5.5
14.1
12.1
11
5.7
2.8
2008
2009
2010
2011
2012
Year
Assumption
Upper Bound
Midpoint
Lower Bound

6 Discussion

6.1 Interpretation of Key Findings

This paper documents a decline in the proportion of workers earning below HK$28 following the introduction of Hong Kong's Statutory Minimum Wage in May 2011. The largest decline in proportion below HK$28 were the workers with No Schooling/Primary education and workers aged 50-64, while tertiary-educated workers and prime-age workers showed less changes. These findings are descriptively consistent with a policy effect concentrated among groups most likely to be affected by a minimum wage.

6.2 Explanations of the Difference in Impacts Between Age and Education Subgroups

One possible explanation for the difference between education and age groups lies in employment composition. As shown in the sample composition analysis (Section 5.4), the employment count for older workers (50-64) increased from 100 in 2008 to 130 in 2012, which is a 30% rise. This suggests that more older workers remained in or entered the labour force during this period. For a group with rapidly expanding employment, year-over-year fluctuations in low-wage shares may be more volatile, making the 2011 decline less historically unique.

In contrast, the employment count for No Schooling/Primary workers declined modestly before 2011 but stabilized afterward. A more stable employment base may allow policy-coincident changes to stand out more clearly from historical fluctuations. This could explain the lack of uniqueness in changes in older age groups across time compared to low-education groups.

6.3 Implications for Policy Evaluation

The implications for policymaking could start with the low-education groups who demonstrated identifiable improvements coinciding with the policy period. Afterwards, the suggestions could explore more complementary measures for complicated subgroups, such as workers aged between 50 and 64.

As discussed above, low-education groups, especially those at the education level "No Schooling/Primary", showed strong improvements in wage distribution. The historically unique improvement not only coincides with the time of the policy but also concentrates around the minimum wage threshold. This suggests that the minimum wage policy, when intertwined with concurrent macroeconomic forces, could be associated with significant improvements in low-education groups' income distribution that go beyond the ordinary economic cycle.

The robustness checks on older age groups revealed that the improvements for workers aged between 50 and 64 were not unique. This group may also be sensitive to other macroeconomic factors which affect employment. Thus, policymakers should realize that the welfare of this group is more embedded in economic cycles and employment fluctuations. To generate a more unique improvement for this group, more targeted policies, such as specific subsidies and anti-age-discrimination legislation, could be considered.

7 Limitations

7.1 Data and measurement limitations

The General Household Survey is a repeated cross-sectional dataset, meaning the individuals in the data will not be the same in each quarter. Although grossing up factor can show the aggregate population in society, tracking the same workers over the 1985 to 2012 period is nearly impossible. Thus, this prevents us from examining if the specific individual workers experienced wage increases and left low wage employment.

7.2 Causal identification

While the proportion of workers earning below HK$28 declined for certain groups from 2010 to 2011, it could not be claimed that it was the policy effect as factors such as economic recovery from 2008 to 2009 global financial crisis can exist and contributed to wage growth, which confounds any possible policy attribution as well.

7.3 Errors in Working Hours Estimation

In the previous estimation of working hours, it was assumed that the workers would work throughout the year every week. However, in reality, workers may enjoy vacations, causing working hours to be zero or much smaller in those weeks with holidays. Our assumption overestimates the number of working hours by assuming equal working hours even in public holiday contexts, which leads to a slight underestimation of hourly wages.

8 Conclusion

This paper documents changes in low wage employment in Hong Kong before and after the introduction of Statutory Minimum Wage (SMW) in May 2011, while focusing on subgroups by education and age.

8.1 Summary of findings

With 1985 to 2012 quarterly data from the General Household Survey, it was found that the proportion of workers earning below HK$28 declined from 7.9% in 2011Q1 to 4.7% in 2011Q3. This motivated a deeper analysis into the changes in heavily exposed groups under this policy.

In further analysis, it was found that, among other education levels, workers with education being “No Schooling / Primary” experienced the largest decline in proportion of workers below the HK$28 threshold, from 23.9% in 2011Q1 to 12.5% in 2011Q3. In terms of age groups, among other age groups, 50-64-year-old workers experienced the largest decline from 12.6% in 2011Q1 to 7.0% in 2011Q3. Those findings serve as evidence of the choice of high-exposure groups under the minimum wage policy.

8.2 Robustness Check Results

To evaluate the uniqueness of the changes in high-exposure groups in 2011, robustness checks were conducted on the high-exposure groups, which compare the change in 2011 with historical fluctuations. The test diagrams suggested that the change in proportion of workers earning below HK$28 per hour who are in the education level “No Schooling / Primary” was unique compared to the past 20 years. In contrast, the same proportion change for the workers in age group between 50 and 64 was relatively less unique considering the size of historical fluctuations where the policy was absent.

Moreover, robustness checks that varied the threshold of focus revealed different patterns across groups. For No Schooling/Primary workers, the decline was concentrated at the HK$28 threshold, but older workers' decline was less concentrated at HK$28. Thus, the decline for older workers was less distinct than for the No Schooling/Primary group.

Additional robustness checks addressed the concerns of errors produced by midpoint estimation and the possibility that the proportion shift was driven by sample composition.

8.3 Final remarks

The reduction of low wage employment proportion coincides with Hong Kong's minimum wage policy. However, the reduction in the perspective of age is not historically unusual.

Due to the underestimation in hourly wages, the actual impact of the policy could be larger than the data shown in this study.

This study showed comparisons of the proportion of low wages before and after policy but are insufficient for causal claims.

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Appendix

Appendix A: Variable Position Mapping Tables

From Q1 1985 to Q4 1992

Item No.	Description	Format	Position	
			From	To
1	Year/ Quarter	9(5)	1	5
2	Household reference no.	9(14)	6	19
3	Relationship to household head **[not available]**	9(2)	20	21
4	Age	9(2)	22	23
5	Sex	9(1)	26	26
6	Educational attainment	9(1)	27	27
7	Marital status	9(1)	28	28
8	Economic activity status	9(1)	29	29
9	Whether being underemployed (for employed persons)	9(1)	30	30
10	Usual place of work **[not available]**	9(1)	31	31
11	Industry (for employed persons excluding underemployed persons)	9(2)	32	33
12	Industry (for underemployed persons)	9(1)	34	34
13	Previous industry (for unemployed persons)	9(1)	35	35
14	Occupation (for employed persons)	9(2)	36	37
15	Previous occupation (for unemployed persons)	9(2)	38	39
16	Hours worked during the seven days before enumeration	9(3)	40	42
17	Hours worked in main employment (for employed persons) **[not available]**	9(3)	43	45
18	Monthly employment earnings (for employed persons excluding underemployed persons)	9(2)	46	47
19	Monthly employment earnings (for underemployed persons)	9(2)	48	49
20	Duration of unemployment (for unemployed persons)	9(1)	50	50

21	Reasons for leaving last job (for unemployed persons with a previous job)	9(1)	51	51
22	Whether being a foreign domestic helper (for employed persons) **[not available]**	9(1)	52	52
23	Grossing-up factor	9(4)v9(12)	53	68
24	Spouse's person serial number **[not available]**	9(2)	69	70
25	Parent's person serial number **[not available]**	9(2)	71	72
26	Year for Completed Studying **[not available]**	9(4)	73	76
27	Length of employment in current job **[not available]**	9(2)	77	78
28	Whether Intend to Work in Same Industry **[not available]**	9(1)	79	79
29	Intend to Work in Which Industry	9(1)	80	80
30	Whether Intend to Work in Same Occupation **[not available]**	9(1)	81	81
31	Intend to Work in Which Occupation **[not available]**	9(2)	82	83
32	Person serial number	9(2)	84	85
33	Hours worked in all secondary employment **[not available]**	9(3)	86	88
34	Monthly earnings from main employment	9(2)	89	90
35	Monthly earnings from all other employment	9(2)	91	92
36	Whether have other employment in the 7 days before enumeration	9(1)	93	93
37	Amount of "Chinese New Year bonus/double pay in last month"	9(2)	94	95

From Q1 1993 to Q4 2000

Item No.	Description	Format	Position	
			From	To
1	Year/ Quarter	9(5)	1	5
2	Household reference no.	9(11)	6	16
3	Relationship to household head	9(2)	17	18
4	Age	9(2)	19	20
5	Sex	9(1)	23	23
6	Educational attainment	9(1)	24	24

7	Marital status	9(1)	25	25
8	Economic activity status	9(1)	26	26
9	Whether being underemployed (for employed persons)	9(1)	27	27
10	Usual place of work **[not available]**	9(1)	28	28
11	Industry (for employed persons excluding underemployed persons)	9(2)	29	30
12	Industry (for underemployed persons)	9(1)	31	31
13	Previous industry (for unemployed persons)	9(1)	32	32
14	Occupation (for employed persons)	9(2)	33	34
15	Previous occupation (for unemployed persons)	9(2)	35	36
16	Hours worked during the seven days before enumeration	9(3)	37	39
17	Hours worked in main employment	9(3)	40	42
18	Monthly employment earnings (for employed persons excluding underemployed persons)	9(2)	43	44
19	Monthly employment earnings (for underemployed persons)	9(2)	45	46
20	Duration of unemployment (for unemployed persons)	9(1)	47	47
21	Reasons for leaving last job (for unemployed persons with a previous job)	9(1)	48	48
22	Whether being a foreign domestic helper (for employed persons)	9(1)	49	49
23	Grossing-up factor	9(4)v9(6)	50	59
24	Spouse's person serial number	9(2)	60	61
25	Parent's person serial number	9(2)	62	63
26	Year for Completed Studying **[not available]**	9(4)	64	67
27	Length of employment in current job **[not available]**	9(2)	68	69
28	Whether Intend to Work in Same Industry	9(1)	70	70
29	Intend to Work in Which Industry	9(1)	71	71
30	Whether Intend to Work in Same Occupation	9(1)	72	72
31	Intend to Work in Which Occupation	9(2)	73	74
32	Person serial number	9(2)	75	76
33	Hours worked in all secondary employment	9(3)	77	79
34	Monthly earnings from main employment	9(2)	80	81

35	Monthly earnings from all other employment	9(2)	82	83
36	Whether have other employment in the 7 days before enumeration	9(1)	84	84
37	Amount of "Chinese New Year bonus/double pay in last month"	9(2)	85	86

From Q1 2001 to Q4 2007

Item No.	Description	Format	Position	
			From	To
1	Year/ Quarter	9(5)	1	5
2	Household reference no.	9(11)	6	16
3	Relationship to household head	9(2)	17	18
4	Age	9(2)	19	20
5	Sex	9(1)	23	23
6	Educational attainment	9(1)	24	24
7	Marital status	9(1)	25	25
8	Economic activity status	9(1)	26	26
9	Whether being underemployed (for employed persons)	9(1)	27	27
10	Usual place of work	9(1)	28	28
11	Industry (for employed persons excluding underemployed persons)	9(2)	29	30
12	Industry (for underemployed persons)	9(1)	31	31
13	Previous industry (for unemployed persons)	9(1)	32	32
14	Occupation (for employed persons)	9(2)	33	34
15	Previous occupation (for unemployed persons)	9(2)	35	36
16	Hours worked during the seven days before enumeration (for employed persons)	9(3)	37	39
17	Hours worked in main employment (for employed persons)	9(3)	40	42
18	Monthly employment earnings (for employed persons excluding underemployed persons)	9(2)	43	44
19	Monthly employment earnings (for underemployed persons)	9(2)	45	46

20	Duration of unemployment (for unemployed persons)	9(1)	47	47
21	Reasons for leaving last job (for unemployed persons with a previous job)	9(1)	48	48
22	Whether being a foreign domestic helper (for employed persons)	9(1)	49	49
23	Grossing-up factor	9(4)v9(6)	50	59
24	Spouse's person serial number	9(2)	60	61
25	Parent's person serial number	9(2)	62	63
26	Year for Completed Studying (for persons aged 15-25 not studying full-time courses)	9(4)	64	67
27	Length of employment in current job (for employee)	9(2)	68	69
28	Whether Intend to Work in Same Industry (for unemployed persons)	9(1)	70	70
29	Intend to Work in Which Industry (for unemployed persons)	9(1)	71	71
30	Whether Intend to Work in Same Occupation (for unemployed persons)	9(1)	72	72
31	Intend to Work in Which Occupation (for unemployed persons)	9(2)	73	74
32	Person's serial number	9(2)	75	76
33	Hours worked in all secondary employment (for employed persons)	9(3)	77	79
34	Monthly earnings from main employment (for employed persons)	9(2)	80	81
35	Monthly earnings from all other employment (for employed persons)	9(2)	82	83
36	Whether have other employment in the 7 days before enumeration (for employed persons)	9(1)	84	84
37	Amount of "Chinese New Year bonus/double pay in last month" (for employed persons)	9(2)	85	86

From Q1 2008 to current year

Item			Position	
No.	Description	Format	From	To
1	Year/ Quarter	9(5)	1	5
2	Household reference no.	9(11)	6	16
3	Relationship to household head	9(2)	17	18

4	Age	9(2)	19	20
5	Sex	9(1)	23	23
6	Educational attainment	9(1)	24	24
7	Marital status	9(1)	25	25
8	Economic activity status	9(1)	26	26
9	Whether being underemployed (for employed persons)	9(1)	27	27
10	Usual place of work	9(1)	28	28
11	Industry (for employed persons excluding underemployed persons)	9(2)	29	30
12	Industry (for underemployed persons)	9(1)	31	31
13	Previous industry (for unemployed persons)	9(1)	32	32
14	Occupation (for employed persons)	9(2)	33	34
15	Previous occupation (for unemployed persons)	9(2)	35	36
16	Hours worked during the seven days before enumeration (for employed persons)	9(3)	37	39
17	Hours worked in main employment (for employed persons)	9(3)	40	42
18	Monthly employment earnings (for employed persons excluding underemployed persons)	9(2)	43	44
19	Monthly employment earnings (for underemployed persons)	9(2)	45	46
20	Duration of unemployment (for unemployed persons)	9(1)	47	47
21	Reasons for leaving last job (for unemployed persons with a previous job)	9(1)	48	48
22	Whether being a foreign domestic helper (for employed persons)	9(1)	49	49
23	Grossing-up factor	9(5)v9(6)	50	60
24	Spouse's person serial number	9(2)	61	62
25	Parent's person serial number	9(2)	63	64
26	Year for Completed Studying (for persons aged 15-25 not studying full-time courses)	9(4)	65	68
27	Length of employment in current job (for employee)	9(2)	69	70
28	Whether Intend to Work in Same Industry (for unemployed persons)	9(1)	71	71
29	Intend to Work in Which Industry (for unemployed persons)	9(1)	72	72
30	Whether Intend to Work in Same Occupation (for unemployed	9(1)	73	73

	persons)			
31	Intend to Work in Which Occupation (for unemployed persons)	9(2)	74	75
32	Person's serial number	9(2)	76	77
33	Hours worked in all secondary employment (for employed persons)	9(3)	78	80
34	Monthly earnings from main employment (for employed persons)	9(2)	81	82
35	Monthly earnings from all other employment (for employed persons)	9(2)	83	84
36	Whether have other employment in the 7 days before enumeration (for employed persons)	9(1)	85	85
37	Amount of "Chinese New Year bonus/double pay in last month" (for employed persons)	9(2)	86	87

Appendix B: Wage Ranges

Code	Description
01	<2,000
02	2,000 – 2,999
03	3,000 – 3,999
04	4,000 – 4,999
05	5,000 – 5,999
06	6,000 – 6,999
07	7,000 – 7,999
08	8,000 – 8,999
09	9,000 – 9,999
10	10,000 – 10,999
11	11,000 – 11,999
12	12,000 – 12,999
13	13,000 – 13,999
14	14,000 – 14,999
15	15,000 – 15,999

16	16,000 – 16,999
17	17,000 – 17,999
18	18,000 – 18,999
19	19,000 – 19,999
20	20,000 – 21,999
21	22,000 – 23,999
22	24,000 – 25,999
23	26,000 – 27,999
24	28,000 – 29,999
25	30,000 – 31,999
26	32,000 – 33,999
27	34,000 – 35,999
28	36,000 – 37,999
29	38,000 – 39,999
30	40,000 – 41,999
31	42,000 – 43,999
32	44,000 – 45,999
33	46,000 – 47,999
34	48,000 – 49,999
35	50,000 – 59,999
36	60,000 – 69,999
37	70,000 – 79,999
38	80,000 – 89,999